University of Calgary
Faculty of Science
Final Exam
PHYS 365

VERSION 01

December 13, 2024, 9:00-11:00 AM Time: 120 minutes

This is a closed-book exam containing a total of 30 equally weighted questions.

ONLY THE FOLLOWING ITEMS ARE ALLOWED ON YOUR DESK DURING THE EXAM:

1. THIS MULTIPLE-CHOICE QUESTION BOOKLET, which includes the exam questions, the formula sheet, and 4 blank pages. This booklet WILL NOT be handed in.

2. A BUBBLE SHEET used to answer multiple-choice questions. This WILL BE handed in.

IMPORTANT: Start by entering your UCID, NAME, COURSE, LECTURE SECTION, and VERSION NUMBER on the top of the bubble sheet.

3. A CALCULATOR. Only a Schulich-approved scientific calculator may be used during the exam.

4. A PEN OR PENCIL, used to black out your answers on the bubble sheet. If you are using a PEN, make sure it is BLACK or BLUE to ensure the scanner reads your answers properly. PENCILS can also be used, but make sure you press firmly when answering so the scanner reads your answer properly.

5. YOUR STUDENT ID CARD, used to check your identity during exam sign-in.

6. Students remaining after 105 minutes must remain seated for the duration of the exam until permitted to leave by the exam supervisor.

1. The sound level made by a whispering person is 30 dB. What is the pressure amplitude of the sound assuming a frequency of 1000 Hz, a density of air of 1.2 $\mathrm{kg/m^3}$, and a speed of sound of 343 m/s.

   A) 9.1×10^{-4} Pa
   B) 1.3×10^{-4} Pa
   C) 2.0×10^{-9} Pa
   D) 3.5×10^{-10} Pa

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2. An acoustic wave of amplitude A is travelling in water, whose impedance is $Z_w = 1.48$ MPa. The wave is incident on a boundary with another liquid with impedance Z_L such that 81% of the incident wave's **power** is transmitted. What is the impedance of the other liquid?

   A) 3.77 MPa
   B) 2.17 MPa
   C) 3.84 MPa
   D) 5.81 MPa

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3. If you set up the resonance for the fifth harmonic on a string, how many nodes and antinodes are in the pattern?

   A) 5 nodes and 6 antinodes
   B) 4 nodes and 5 antinodes
   C) 6 nodes and 5 antinodes
   D) 5 nodes and 4 antinodes

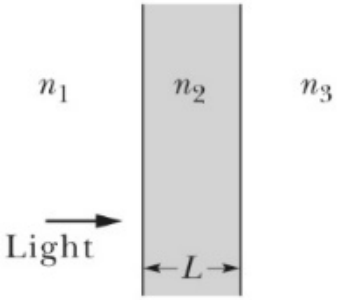
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4. An organ pipe with one end open and the other closed is operating at one of its resonant frequencies. The open and closed ends are, respectively:

   A) pressure node, pressure node
   B) pressure node, displacement node
   C) displacement antinode, pressure node
   D) displacement node, displacement node

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5. The figure below shows a thin film with different materials on either side, where $n_1 = 1.30$, $n_2 = 1.50$, and $n_3 = 1.40$. A ray of light with a wavelength in vacuum of 600 nm is incident normal to the film. What is the second nonzero smallest value of the film thickness L that produces **destructive** interference in the reflected waves?



A) 400 nm
B) 200 nm
C) 100 nm
D) 300 nm

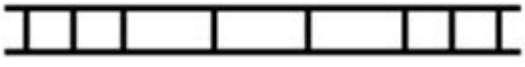
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6. In a Young's double-slit experiment, the slit separation is doubled. To maintain the same fringe spacing on the screen, the screen-to-slit distance D must be changed to

A) $D/2$
B) $D\sqrt{2}$
C) $2D$
D) $4D$

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7. Monochromatic plane waves of light are incident normally on a single slit. Which one of the four figures below most correctly shows the diffraction pattern observed on a distant screen?

A) 
B) 
C) 
D) 

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8. A phasor $\tilde{A}$ expressed in cartesian form has the following real and imaginary components:

$$Re[\tilde{A}] = -7.25 \text{ mm}$$
$$Im[\tilde{A}] = +4.38 \text{ mm}$$

What is the phase constant of the phasor?

A) -0.543 rad
B) -1.03 rad
C) 2.11 rad
D) 2.60 rad

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9. A wave pulse is travelling through a medium in the $-x$-direction at a speed of 20 m/s. Which of the following functions could possibly describe the propagating pulse?

A) $f(x,t) = 20(x+t)^2$

B) $f(x,t) = 5e^{-(40t+2x)^2}$

C) $f(x,t) = \frac{50}{60t-3x}$

D) $f(x,t) = 200t^2 + 10x^2$

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10. The position of an object undergoing simple harmonic motion is given by:

$$x(t) = A\sin(\omega t + \pi/4)$$

Which expression below gives the acceleration of the object?

A) $a(t) = -A\omega^2(\sin(\omega t) + \cos(\omega t))$

B) $a(t) = A\omega^2\cos(\omega t + \pi/4)$

C) $a(t) = \frac{A\omega^2}{\sqrt{2}}(\sin(\omega t) + \cos(\omega t))$

D) $a(t) = A\omega^2\cos(\omega t + 3\pi/4)$

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11. A simple pendulum is made from a ball of mass m hanging from a string of length ℓ and when it is set swinging on Earth it is observed to have period T. The pendulum is then taken to the moon where the acceleration due to gravity is $1/6$ that on Earth. What is the period of the pendulum on the moon?

A) It is $\sqrt{6}$ times longer than on Earth
B) It is 6 times longer than on Earth
C) It is $1/6$ as long as on Earth
D) It is the same as on Earth

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12. The superposition of two waves of equal amplitude and opposite phase constants,

$$y_1(x,t) = A\sin(kx - \omega t + \phi_0)$$
$$y_2(x,t) = A\sin(kx - \omega t - \phi_0)$$

yields the resultant wave $y_r(x,t) = \sqrt{3}A\sin(kx - \omega t)$. What is the phase constant ϕ_0 that produces this resultant wave?

A) $\pi/6$
B) $3\pi/4$
C) $\pi/3$
D) $\pi/8$

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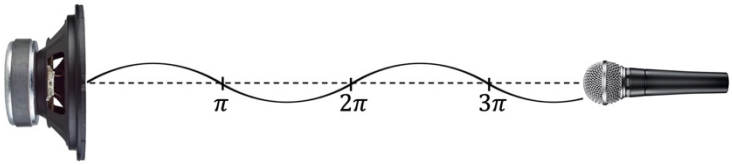
13. Noise-cancelling headphones detect a sound wave of frequency 440 Hz. It takes the headphones 2.00 ms to register the signal and emit a wave with the same amplitude and frequency. Taking the phase constant of the initial wave to be zero, what should the phase constant of the emitted wave be to cancel the noise?

A) -0.76π
B) 0.76π
C) 1.76π
D) -0.24π

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14. A speaker and a microphone are placed 1.00 m apart. The frequency of the emitted wave is 600 Hz and the phase difference between the emitted and detected wave is $\Delta\phi = 3.75\pi$, as shown in the diagram. What is the temperature of the air?
(Hint: air is made of a mixture of N_2 and O_2 with a molecular mass of 28.8 g/mol)

A) $-20°C$
B) $+20°C$
C) 212 K
D) 273 K



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15. Within the first few kilometers above sea level, the atmospheric pressure p and air density ρ vary with altitude h according to

$$p(h) = p_0\left(1 - \frac{h}{\alpha}\right) \qquad \text{and} \qquad \rho(h) = \rho_0\left(1 - \frac{h}{\beta}\right)$$

where $\alpha = 8.4$ km, $\beta = 10.4$ km, and p_0 and ρ_0 are the air pressure and density at sea level. How does the acoustic impedance of air change with altitude near Earth's surface?

Hint: $\Delta\rho = \Delta\left(\frac{M}{V}\right) = \frac{\partial}{\partial V}\left(\frac{M}{V}\right)\Delta V = \left(-\frac{M}{V^2}\right)\Delta V = -\frac{\rho}{V}\Delta V$, so you can use $\frac{\Delta V}{V} = -\frac{\Delta\rho}{\rho}$.

A) It is approximately constant: $Z(h) = Z_0$

B) It decreases slowly with h: $Z(h) = Z_0\sqrt{1 - \frac{h}{\beta}}$

C) It decreases linearly with h: $Z(h) = Z_0\left(1 - \frac{h}{\beta}\right)$

D) It decreases quadratically with h: $Z(h) = Z_0\left(1 - \frac{h}{\beta}\right)^2$

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16. A laser beam with wavelength 720 nm is shone on a single slit aperture. On the detection screen 2.00 m away, the first dark fringe of the diffraction pattern is located 1.83 cm away from the central maximum. What is the width of the slit?

A) 118 μm
B) 39.3 μm
C) 59.0 μm
D) 78.7 μm

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17. The Hubble Space Telescope is imaging distant galaxies at a wavelength of 550 nm. The diameter of the telescope's objective is 2.40 m. What is the smallest angular resolution of the Hubble Space Telescope at this wavelength?

A) 1.602×10^{-5} rad
B) 3.204×10^{-5} rad
C) 2.796×10^{-7} rad
D) 5.592×10^{-7} rad

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18. The light corresponding to the Lyman spectral series of hydrogen is observed from a distant star and it appears to have a slightly longer wavelength than normal. Which statement below must be true?

A) Earth is moving toward the star.
B) The star is moving away from Earth.
C) The star is moving toward Earth.
D) The distance between Earth and the star remains the same.

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19. A free electron in motion along the x-axis has a localized wave function. If it enters a region of space where its potential energy increases:

A) its wavelength will increase.
B) its kinetic energy will increase.
C) its wave number will increase.
D) its total energy will decrease.

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20. Three particles with masses m_1, $m_2 = 2m_1$, and $m_3 = 3m_1$ are confined in the same one-dimensional space. Measurements of the speeds of the particles are made while keeping the three particles confined in the space. Which particle can have its speed measured with the least amount of uncertainty?

A) Particle 1 with mass m_1
B) Particle 2 with mass m_2
C) Particle 3 with mass m_3
D) All particles have the same minimum uncertainty

Questions 21 - 23 are related to the following scenario:

An electron is confined to a one-dimensional infinite potential well of width $L = 0.100$ nm.

21. If the electron is in the second excited state, what wavelengths of photons could this system emit?

A) 4.1 nm only
B) 4.1 nm, and 6.6 nm only
C) 4.1 nm, 6.6 nm, and 11 nm only
D) 11 nm only

22. If the electron is in its ground state, what is the probability of finding it between $x = L/8$ and $x = L/4$?

A) 7.84%
B) 14.1%
C) 3.92%
D) 25.0%

23. Which equation describes the wave function if the electron is in the third excited state?

A) $\psi(x) = \left(1.99 \times 10^{10}\ \mathrm{m}^{-\frac{1}{2}}\right) \sin\left((1.26 \times 10^{11}\ \mathrm{m}^{-1})x\right)$
B) $\psi(x) = \left(1.41 \times 10^{5}\ \mathrm{m}^{-\frac{1}{2}}\right) \sin\left((9.42 \times 10^{10}\ \mathrm{m}^{-1})x\right)$
C) $\psi(x) = \left(1.99 \times 10^{10}\ \mathrm{m}^{-\frac{1}{2}}\right) \sin\left((9.42 \times 10^{10}\ \mathrm{m}^{-1})x\right)$
D) $\psi(x) = \left(1.41 \times 10^{5}\ \mathrm{m}^{-\frac{1}{2}}\right) \sin\left((1.26 \times 10^{11}\ \mathrm{m}^{-1})x\right)$

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24. A hydrogen atom is in its second excited state. To what state would it jump if it were to absorb a photon with the longest possible wavelength that it could?

A) $n = 2$
B) $n = 3$
C) $n = 4$
D) $n = 5$

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25. The target of a photoelectric experiment is made from an unknown metal. When the frequency of the incident light is $f_1 = 3.00 \times 10^{15}$ Hz, the stopping voltage is measured to be $V_1 = 4.57 \times 10^{19}$ V. What metal is the target made from?

A) Silver $\Phi_{\mathrm{Ag}} = 4.46$ eV
B) Gold $\Phi_{\mathrm{Au}} = 5.10$ eV
C) Aluminum $\Phi_{\mathrm{Al}} = 4.06$ eV
D) Copper $\Phi_{\mathrm{Cu}} = 4.53$ eV

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26. A hydrogen atom initially at rest absorbs a photon with frequency $f = 2.923$ GHz. What is the speed of the hydrogen atom after the absorption?

A) 7.092×10^{-3} m/s
B) 4.868×10^{-9} m/s
C) 3.868×10^{-6} m/s
D) 4.424×10^{16} m/s

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27. A photon travelling in the x-direction Compton scatters off an electron. Of the scattering angles listed below, which one corresponds to the outgoing photon having the largest wavelength shift compared to the initial photon?

A) $\pi/3$
B) $3\pi/2$
C) $2\pi/3$
D) $\pi/4$

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28. Which of the following statements about an electron microscope is TRUE?

A) The resolving power of an electron microscope is determined exclusively by the mass of the electrons.
B) One can increase the resolving power of an electron microscope by accelerating the electrons more.
C) One can increase the resolving power of an electron microscope if instead of electrons positrons are used.
D) The resolving power of an electron microscope will decrease if instead we use protons.

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29. An electron beam is composed of electrons moving at a speed of $v = 8.54 \times 10^4$ m/s, which is slow enough compared to c that relativistic effects can be ignored. The diffraction picture produced by the electron beam is identical to that of a beam of x-ray photons. What is the frequency of the x-ray beam?

A) 1.24×10^{17} Hz
B) 2.27×10^{18} Hz
C) 3.52×10^{16} Hz
D) 5.01×10^{16} Hz

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30. A bright region of a certain diffraction picture has an intensity that is 100 times that of a darker region of the same picture: $I_d = I_b/100$. What is the ratio of the probability density of a single photon hitting the bright vs the dark regions, $\eta = P_b/P_d$?

A) $\eta = 100$
B) $\eta = 10000$
C) $\eta = 10^{-2}$
D) $\eta = 1$

PHYS 365 Final Exam Formula Sheet

OSCILLATIONS

$x(t) = x_m \cos(\omega t + \phi)$	$\omega = 2\pi f = \frac{2\pi}{T}$	$\omega = \sqrt{\frac{k}{m}}$	$\omega = \sqrt{\frac{g}{\ell}}$	$T = 2\pi\sqrt{\frac{\ell}{g}}$

WAVES

$\frac{\partial^2 y}{\partial^2 t} = v^2 \frac{\partial^2 y}{\partial x^2}$	$y(x,t) = f(x \pm vt)$	$y(x,t) = y_m \cos(kx \pm \omega t + \phi)$	
$v = \frac{\omega}{k} = f\lambda$	$k = \frac{2\pi}{\lambda}$	$v = \frac{c}{n}$	$\lambda = \frac{\lambda_0}{n}$
$v = \sqrt{\frac{\tau}{\mu}}$	$v = \sqrt{\frac{\gamma RT}{M}}$	$v = \sqrt{\frac{Y}{\rho}}$	$v = \sqrt{\frac{B}{\rho}}$
$Z = \sqrt{\mu\,\tau} = \frac{\tau}{v} = \mu v$	$Z = \sqrt{\rho B} = \rho v$	$P_{avg} = \frac{1}{2}\mu v \omega^2 y_m^2$	$\Delta p_m = (v\rho\omega)s_m$
$\Delta p = \Delta p_m \sin(kx - \omega t)$	$I = \frac{1}{2}\rho v \omega^2 s_m^2$	$I = \frac{P}{4\pi R^2}$	$I = \frac{P}{A}$
$\beta = (10\ dB)\log_{10}\left(\frac{I}{I_0}\right)$	$\Delta\beta = (10\ dB)\log_{10}\left(\frac{I_2}{I_1}\right)$	$I(r_2) = \left(\frac{r_1}{r_2}\right)^{n-1} I(r_1)e^{-\alpha\Delta r}$	
$R = \frac{Z_1 - Z_2}{Z_1 + Z_2}$	$T = \frac{2Z_1}{Z_1 + Z_2}$	$B = -\frac{\Delta p}{\left(\frac{\Delta V}{V}\right)}$	$Y = \frac{\left(\frac{F}{A}\right)}{\left(\frac{\Delta L}{L}\right)}$
$R_e = R^2$	$T_e = \frac{Z_2}{Z_1}T^2$	$\Delta p = -B\frac{\partial s}{\partial x}$	

WAVE INTERFERENCE

$y'(x,t) = \left[2y_m \cos\left(\frac{1}{2}\phi\right)\right]\sin\left(kx - \omega t + \frac{1}{2}\phi\right)$		$y'(x,t) = [2y_m \sin(kx)]\cos(\omega t)$
$\Delta\phi = \frac{\Delta L}{\lambda}2\pi$	$f_{beat} = f_1 - f_2$	$f' = f\frac{v \pm v_D}{v \pm v_S}$
Constructive interference $\frac{\Delta L}{\lambda} = 0,\ 1,\ 2, \ldots$		Destructive interference $\frac{\Delta L}{\lambda} = 0.5,\ 1.5,\ 2.5, \ldots$
Taut string or pipe open at both ends $f_n = \frac{v}{\lambda_n} = \frac{nv}{2L}$ for $n = 1,\ 2,\ 3, \ldots$		Pipe one end closed, one end open $f_n = \frac{v}{\lambda_n} = \frac{nv}{4L}$ for $n = 1,\ 3,\ 5, \ldots$
$d\sin(\theta) = m\lambda$ for $m = 0, 1, 2, \ldots$ (maxima-bright fringes)		
$d\sin(\theta) = \left(m + \frac{1}{2}\right)\lambda$ for $m = 0, 1, 2, \ldots$ (minima-dark fringes)		

$2t = m\frac{\lambda_0}{n}, \quad m = 1, 2, 3 \ldots$		$2t = \left(m - \frac{1}{2}\right)\frac{\lambda_0}{n}, \quad m = 1, 2, 3 \ldots$	
Reflection	Reflection phase shift	$a\sin(\theta) = m\lambda$ for $m = 1, 2, 3, \ldots$ (minima)	
Off lower index	0		
Off higher index	0.5 wavelength		
$I = 4I_0\cos^2\left(\frac{1}{2}\phi\right)$	$\phi = \frac{2\pi d}{\lambda}\sin(\theta)$	$\theta_R = \sin^{-1}\left(\frac{1.22\lambda}{d}\right)$	
MODERN PHYSICS			
$F_e = \frac{1}{4\pi\epsilon_0}\frac{e^2}{r^2}$	$U = \frac{1}{4\pi\epsilon_0}\frac{e^2}{r}$	$U = qV$	$K = \frac{1}{2}mv^2$
$E = hf$	$p = \frac{h}{\lambda}$	$hf = K_{max} + \Phi$	$a_c = \frac{v^2}{r}$
$\Delta\lambda = \frac{h}{mc}(1 - \cos\phi)$	$\frac{1}{\lambda} = R\left(\frac{1}{n_{low}^2} - \frac{1}{n_{high}^2}\right)$	$K_{max} = eV_{stop}$	$R = \frac{P}{hf}$
$\Delta x\Delta p_x \geq \frac{h}{2\pi}$	$a_0 = \frac{\epsilon_0 h^2}{\pi m_e e^2}$	$E_n = -\frac{m_e e^4}{8\epsilon_0^2 h^2 n^2}$	$E_n = \frac{-13.6\text{ eV}}{n^2}$
$E_n = \frac{h^2}{8m_e L^2}n^2$	$\psi_n(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi}{L}x\right)$	$P(x_1, x_2) = \int_{x_1}^{x_2}\lvert\psi(x)\rvert^2 dx$	
$P(r) = \frac{4}{a_0^3}r^2 e^{\frac{-2r}{a_0}}$	$\psi(r) = \frac{1}{\sqrt{\pi}a_0^{3/2}}e^{\frac{-r}{a_0}}$	$T \approx e^{-2bL}$ $b = \sqrt{8\pi^2 m_e(U_b - E)/h^2}$	
$\frac{d^2\psi}{dx^2} + k^2\psi = 0$, with $k = \frac{2\pi\sqrt{2m_e(E-U)}}{h}$		$\frac{d^2\psi}{dx^2} + \frac{8\pi^2 m_e}{h^2}\big(E - U(x)\big)\psi = 0$	
CONSTANTS			
$c = 3.00 \times 10^8\ \frac{\text{m}}{\text{s}}$	$R = 8.314\ \frac{\text{J}}{\text{mol}\cdot\text{K}}$	$M_{air} = 28.8\ \frac{\text{g}}{\text{mol}}$	$g = 9.81\ \frac{\text{m}}{\text{s}^2}$
$v_{sound\ at\ 20℃} = 343\ \frac{\text{m}}{\text{s}}$	$T_K = T_{℃} + 273.15$	$\gamma_{monoatomic} = \frac{5}{3}$	
$I_0 = 1 \times 10^{-12}\ \frac{\text{W}}{\text{m}^2}$	$1\text{ atm} = 1.01 \times 10^5\text{ Pa}$	$\gamma_{diaatomic} = \frac{7}{5}$	
$m_e = 9.109 \cdot 10^{-31}\text{ kg}$		$m_p = 1.67 \cdot 10^{-27}\text{ kg}$	
$a_0 = 0.5292 \cdot 10^{-10}\text{ m}$		$1\ u = 1.66054 \cdot 10^{-27}\text{ kg}$	

$h = 6.63 \cdot 10^{-34}\ \mathrm{J \cdot s} = 4.136 \cdot 10^{-15}\ \mathrm{eV \cdot s}$	$\epsilon_0 = 8.854 \cdot 10^{-12}\ \dfrac{\mathrm{C^2}}{\mathrm{N \cdot m^2}}$
$\sigma = 5.67 \cdot 10^{-8}\ \dfrac{\mathrm{W}}{\mathrm{m^2 \cdot K^2}}$	$e = 1.602 \cdot 10^{-19}\ \mathrm{C}$
$k_B = 1.381 \cdot 10^{-23}\ \dfrac{\mathrm{J}}{\mathrm{K}} = 8.617 \cdot 10^{-5}\ \dfrac{\mathrm{eV}}{\mathrm{K}}$	Rydberg constant $R = 1.09737 \cdot 10^{7}\ \mathrm{m^{-1}}$
MATHEMATICAL FORMULAE	
$\cos(\alpha) \pm \cos(\beta) = 2\cos\left(\frac{1}{2}(\alpha \pm \beta)\right)\cos\left(\frac{1}{2}(\alpha \mp \beta)\right)$	
$\sin(\alpha) \pm \sin(\beta) = 2\sin\left(\frac{\alpha \pm \beta}{2}\right)\cos\left(\frac{\alpha \mp \beta}{2}\right)$	
$\cos(\alpha \pm \beta) = \cos(\alpha)\cos(\beta) \mp \sin(\alpha)\sin(\beta)$	$\tilde{A} = Ae^{j\theta} = A(\cos(\theta) + j\sin(\theta))$
$\sin(\alpha \pm \beta) = \sin(\alpha)\cos(\beta) \pm \cos(\alpha)\sin(\beta)$	$x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
$\sin(-\alpha) = -\sin(\alpha)$	$\cos\left(\alpha \pm \frac{\pi}{2}\right) = \mp\sin(\alpha)$
$\cos(-\alpha) = \cos(\alpha)$	$\sin\left(\alpha \pm \frac{\pi}{2}\right) = \pm\cos(\alpha)$
$\int \cos(ax) = \dfrac{\sin(ax)}{a} + c$	$\int \sin(ax) = -\dfrac{\cos(ax)}{a} + c$
$\int \sin^2(ax)\,dx = \dfrac{x}{2} - \dfrac{\sin(2ax)}{4a} + c$	$\int \cos^2(ax)\,dx = \dfrac{x}{2} + \dfrac{\sin(2ax)}{4a} + c$
$\sin^2\theta = \dfrac{1}{2}(1 - \cos(2\theta))$	$\cos^2\theta = \dfrac{1}{2}(1 + \cos(2\theta))$